	JSPM University Pune **Faculty of Science and Technology** **School of Computational Sciences** **SY B. Tech. (CSE&IT)**
Experiment No: 1	**Date: / /**
Aim: Write an ALP to perform arithmetic operations using 8051	

Aim of Practical:

1. To understand and implement basic arithmetic operations using the 8051 microcontroller.
2. To write an Assembly Language Program (ALP) for performing addition, subtraction, multiplication, and division.
3. To store and display the results of arithmetic operations.
4. To develop skills in 8051 programming using assembly language.

Equipments Required:

- 8051 Microcontroller Kit (or Keil uVision Simulator)
- Personal Computer with Keil Software
- Power Supply
- USB to Serial Cable (if using hardware)

Theory:

The 8051 microcontroller is an 8-bit microcontroller widely used in embedded systems. It supports arithmetic operations such as addition, subtraction, multiplication, and division..

The **8051 microcontroller** has an **8-bit ALU (Arithmetic Logic Unit)** that can perform basic arithmetic operations. The following registers are commonly used:

- **A (Accumulator):** Used for arithmetic and logical operations.
- **B Register:** Used in multiplication and division.
- **R Registers (R0-R7):** Used for temporary storage

Algorithm :-

Assembly Language Programs (ALP)

1. Addition of Two Numbers

Program	Comments

2. Subtraction of Two Numbers

Program	Comments

3. Multiplication of Two Numbers

Program	Comments

4. Division of Two Numbers

Program	Comments

Conclusion:

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Experiment No: 2	**Date: / /**
Aim: Write an assembly language program to perform logical operations using 8051	

Aim of the Practical:

1. Understand the **logical operations** in the **8051 microcontroller**.
2. Learn how to use **ANL (AND), ORL (OR), XRL (XOR), and CPL (NOT)** instructions in **Assembly Language**.
3. Perform **bitwise operations** on 8-bit data using **logical instructions**.
4. Observe the effect of logical operations on different binary values.
5. Implement logical operations in **Keil uVision** or **8051 hardware** for simulation and execution.

Equipments Required:

- 8051 Microcontroller Kit (or Keil uVision Simulator)
- Personal Computer with Keil Software
- Power Supply
- USB to Serial Cable (if using hardware)

Theory:

Logical operations are essential in digital systems for decision-making and bitwise manipulation. The **8051 microcontroller** provides instructions for performing logical operations on the **accumulator (A register)** and other registers or immediate values.

Logical Instructions in 8051:

Operation	Instruction	Description
AND	**ANL A, Rn**	AND A with register Rn
OR	**ORL A, Rn**	OR A with register Rn
XOR	**XRL A, Rn**	XOR A with register Rn
NOT (Complement)	**CPL A**	Complement (invert) A

Algorithm:-

1. *AND Operation (Bitwise AND)*

Program	Comments

2. OR Operation (Bitwise OR)

Program	Comments

3. XOR Operation (Bitwise XOR)

Program	Comments

4. NOT Operation (Bitwise Complement)

Program	Comments

Conclusion:

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Experiment No: 3	**Date: / /**
Aim: Write an ALP to move a block of n bytes of data from the source (20h) to the destination (40h) using Internal RAM.	

Aim of the Practical:

1. Understand **block transfer operations** in **8051 Assembly Language**.
2. Learn to use **register indirect addressing** for memory operations.
3. Implement a **loop structure** for handling multiple data transfers.
4. Perform data transfer using **internal RAM** of the 8051 microcontroller.
5. Simulate and test the program using **Keil uVision** or **real hardware**.

Equipments Required:

- 8051 Microcontroller Kit (or Keil uVision Simulator)
- Personal Computer with Keil Software
- Power Supply
- USB to Serial Cable (if using hardware)

Theory:

The **8051 microcontroller** has an **internal RAM of 128 bytes (00H – 7FH)**. The block transfer operation involves copying multiple bytes from one memory location to another.

Algorithm :-

Step 1: Initialize Registers

- Set R0 = 20H (Source Address)
- Set R1 = 40H (Destination Address)
- Set R2 = n (Number of Bytes to Transfer)

Step 2: Check if R2 = = 0

- If R2 = 0, go to **Step 6** (End).
- Otherwise, proceed to **Step 4**.

Step 3: Move Data Byte

- Load byte from source address @R0 into Accumulator A.
- Store the value of A into destination address @R1.

Step 4: Update Pointers and Counter

- Increment R0 (Move to next source byte).
- Increment R1 (Move to next destination byte).
- Decrement R2 (Reduce count).
- Repeat **Step 2**.

Step 5: Stop

Code :-

Program	Comments

Conclusion:

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Experiment No: 4	**Date: / /**
Aim: Write an assembly language program to exchange the source block starting with address 20h (Internal RAM) containing N (05) bytes of data with the destination block starting with address 40h (Internal RAM).	

Aim of the Practical:

- To understand the data exchange process in 8051 microcontroller internal RAM.
- To implement block exchange using assembly language instructions.
- To learn memory addressing and data movement in assembly programming.

Equipments Required:

- 8051 Microcontroller Kit (or Keil uVision Simulator)
- Personal Computer with Keil Software
- Power Supply
- USB to Serial Cable (if using hardware)

Theory:

he 8051 microcontroller has 128 bytes of internal RAM (00H-7FH). The data exchange process involves swapping bytes between two memory locations. This requires using registers, indirect addressing, and looping structures to process multiple bytes efficiently. The XCH instruction is useful for exchanging data between registers and memory locations.

Algorithm:

Code :

Program	Comments

Conclusion:

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Experiment No: 5	**Date: / /**
Aim: Write an assembly language program to add the byte in the RAM at 34h and 35h, and store the result in the register R5 (LSB) and R6 (MSB), using Indirect Addressing Mode.	

Aim of the Practical:

1. Understand **addition operations** in **8051 Assembly Language**.
2. Learn how to use **Indirect Addressing Mode** for memory operations.
3. Implement **carry handling** for **16-bit addition**.
4. Perform arithmetic operations using **internal RAM of the 8051 microcontroller**.
5. Simulate and test the program using **Keil uVision** or **real hardware**.

Equipments Required:

- 8051 Microcontroller Kit (or Keil uVision Simulator)
- Personal Computer with Keil Software
- Power Supply
- USB to Serial Cable (if using hardware)

Theory:

The **8051 microcontroller** has **128 bytes of internal RAM (00H – 7FH)**. **Indirect Addressing Mode** is used to access memory locations using **registers as pointers** (R0 or R1).

Addition of two 8-bit numbers results in a 16-bit sum, which requires two registers to store the result:

- **Lower byte (LSB) is stored in R5**
- **Higher byte (MSB) is stored in R6**

The **Carry Flag (CY)** is used to store the **MSB (higher byte)** of the result.

Algorithm :

Code :-

Program	Comments

Conclusion: